

“PLAY STORE INSIGHT ENGINE”

**24ADI204 - DATA SCIENCE AND
VISUALISATION**

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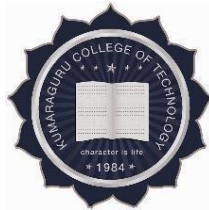
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*in partial fulfilment for the
award of the degree of*

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



KUMARAGURU COLLEGE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

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BONAFIDE CERTIFICATE

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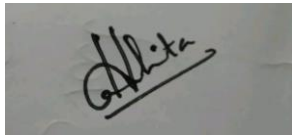
DECLARATION

I affirm that the project work titled “**PLAY STORE INSIGHT ENGINE**” being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

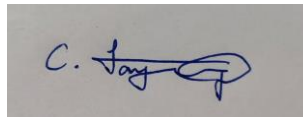
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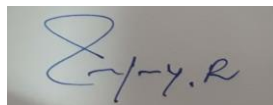
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KUMARAGURU COLLEGE OF TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

FINAL PROJECT REPORT

Play Store Insight Engine
Google Play ecosystem

Course Name: Data Science and Visualization

Course Code: 24ADI204

Academic Year: 2025 – 2026

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Project Overview

This report presents a comprehensive Exploratory Data Analysis (EDA) and preprocessing pipeline developed for the Google Play Store dataset containing over 2.3 million app records. The project was executed as part of the DATA-VIZ HACK-A-BIT initiative under the course Data Science and Visualization (24ADI204) at Kumaraguru College of Technology.

Problem Statement

The Google Play Store hosts millions of applications across diverse categories. Understanding what factors drive user satisfaction (ratings) and app adoption (installs) is critical for developers and product managers. This project aims to identify and quantify relationships between app metadata, monetization strategy, and user engagement metrics.

Dataset

Metric	Value
Source	Kaggle – Google Play Store Dataset
File	Google-Playstore.csv
Total Records	2,312,944 rows
Features	24 columns
Missing Values (raw)	1,305,751
Post-cleaning Records	2,312,939 rows 21 columns
Post-cleaning Missing	0

Key Findings

- App ratings are generally high (median ≈ 4.2) for rated apps across all categories.
- Install-related variables are heavily right-skewed; log transformation is essential.
- Paid apps have marginally higher average ratings (4.14) vs free apps (4.08), but the difference is statistically minor.
- Monetization features (ads, in-app purchases) show no measurable impact on median ratings.
- Rating Count and Installs share a moderate correlation ($r = 0.55$) — scale drives engagement.
- PCA with 11 components explains $\sim 90\%$ of the total variance in the 15-feature engineered matrix.
- Education leads all categories with 241,090 apps; Music & Audio has the highest average rating (4.29).

"So What?" — Final Insights

App quality and user experience matter more than price or monetization style. Since 98% of apps are free, developers should focus on ad quality and in-app purchase design rather than paywalls. Because simple metadata explains very little of rating variance, richer signals — review text, retention rates, crash metrics — are needed for robust predictive models. PCA reveals significant feature redundancy, enabling leaner, faster ML pipelines.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
ML	Machine Learning
KCT	Kumaraguru College of Technology
KSI	Kumaraguru School of Innovation
CSV	Comma-Separated Values
API	Application Programming Interface
IQR	Interquartile Range
NLP	Natural Language Processing
DSV	Data Science and Visualization

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1.1 Background of the Problem

The Google Play Store is the world's largest app marketplace, hosting over 3 million applications across hundreds of categories. Every app developer and product team seeks to understand what drives user satisfaction — reflected in star ratings — and adoption — reflected in install counts. Despite enormous volumes of data generated daily, the relationship between app characteristics (category, pricing, monetization model, size, update frequency) and performance metrics (rating, installs) remains poorly understood at scale.

The dataset analysed in this project, sourced from Kaggle, contains 2.3 million app records with 24 features spanning metadata, user feedback proxies, pricing, and monetization signals. Applying rigorous EDA and preprocessing to this dataset provides a foundation for data-driven product decisions and predictive modelling.

1.2 Importance of the Study

This study is significant for several stakeholders:

- **App Developers:** Understand which categories are saturated and what quality signals matter.
- **Product Managers:** Identify optimal monetization strategies based on engagement data.
- **Investors:** Evaluate app portfolio performance using data-driven benchmarks.
- **Data Scientists:** Build a clean, feature-engineered dataset ready for ML pipelines.

1.3 Objectives of the Project

1. Perform end-to-end EDA on app metadata and user-feedback proxies.
2. Detect and handle missing values, duplicates, and format inconsistencies.
3. Generate actionable univariate, bivariate, and multivariate insights.
4. Build a cleaned, encoded feature set suitable for machine learning.
5. Apply PCA to reduce dimensionality while preserving 90%+ of variance.
6. Develop an interactive visualization dashboard for stakeholder communication.

1.4 Scope of the Project

The project covers the complete data lifecycle from raw ingestion to EDA, preprocessing, feature engineering, and dimensionality reduction. It does not include supervised model training or deployment, though the processed dataset is structured specifically to support downstream ML tasks. The analysis focuses on structured metadata fields; app review text (NLP) is identified as future scope.

2.1 Source of Dataset

The primary dataset used is the **Google Play Store Apps** dataset, available on Kaggle. The file Google-Playstore.csv contains app-level records scraped from the Google Play Store. This is a widely-used benchmark dataset for EDA and recommendation research.

2.2 Dataset Size

Metric	Value
Total Rows	2,312,944
Total Columns	24
Duplicate Rows	0
Total Missing Values	1,305,751
File Format	CSV (Comma-Separated Values)

Table 2.1 – Raw Dataset Profile

2.3 Description of Features

The dataset contains 24 columns spanning four broad categories:

Feature	Type	Description
App Name	Categorical	Name of the application
Category	Categorical	App category (48 unique values)
Rating	Numeric	Average user rating (0–5)
Rating Count	Numeric	Total number of user ratings
Installs	Ordinal Text	Install count bracket (e.g., "1,000,000+")
Minimum Installs	Numeric	Lower bound of install count
Maximum Installs	Numeric	Upper bound of install count
Free	Boolean	Whether the app is free (True/False)
Price	Numeric	App price in USD
Currency	Categorical	Currency code (mostly USD)
Size	Text	App size (e.g., "10M", "varies with device")
Minimum Android	Text	Minimum Android OS version required
Developer Id	Categorical	Unique developer identifier
Developer Website	Text/URL	Developer website URL (high missing)
Developer Email	Text	Developer contact email
Released	Date	App release date

Last Updated	Date	Date of most recent update
Content Rating	Categorical	Target audience rating (6 groups)
Privacy Policy	URL	Link to privacy policy (high missing)
Ad Supported	Boolean	Whether app shows ads
In App Purchases	Boolean	Whether app offers in-app purchases
Editors Choice	Boolean	Whether app has Editor's Choice badge
Scraped Time	Datetime	Timestamp of data collection

Table 2.2 – Feature Description

2.4 Initial Data Issues

Issue Type	Column(s) Affected	Missing Count
High missingness (>30%)	Developer Website	760,835
High missingness (>15%)	Privacy Policy	420,953
Moderate missingness	Released	71,053
Low missingness	Rating, Rating Count	22,883 each
Very low missingness	Minimum Android	6,530
Minimal missingness	Size, Currency, Installs	196, 135, 107
Format inconsistency	Installs (text with commas/+)	—
Skewness	Installs, Price, Rating Count	—

Table 2.3 – Initial Data Issues

Insight: Developer Website (33%) and Privacy Policy (18%) have very high missing rates. These columns offer low analytical value and were dropped during preprocessing. Other missing values were handled via median imputation, preserving data volume.

3.1 Data Loading and Inspection

The dataset was loaded using pandas into a DataFrame. Initial inspection included shape verification, column data type classification, and summary statistics. Numeric columns (Rating, Price, Installs) were distinguished from categorical (Category, Content Rating) and date-type columns (Released, Last Updated).

3.2 Handling Missing Values

Missing values were treated with the following methodology:

Column	Missing Count	Strategy	Justification
Developer Website	760,835	Dropped column	Low analytical value; >33% missing
Privacy Policy	420,953	Dropped column	URL-type, not used in modelling
Released	71,053	Mode imputation	Temporal feature; mode = 2020
Rating	22,883	Median imputation	Rating $\approx$ 4.2; median preserves distribution
Rating Count	22,883	Median imputation	Skewed variable; median preferred
Minimum Android	6,530	Mode imputation	Categorical OS version
Size	196	Median imputation	Numerical size field after parsing
Currency / Installs	107–135	Mode/Median imputation	Minimal impact; safe to impute

Table 3.1 – Missing Value Treatment

3.3 Outlier Detection (IQR Method)

Outlier detection was performed using the Interquartile Range (IQR) method for numeric columns. Given the extreme right-skew in install and rating count data, outliers were **retained but transformed** using log1p rather than removed. Removal would result in loss of high-performing apps that are genuinely significant data points.

IQR Formula: Outlier if $x < Q1 - 1.5 \times IQR$ or $x > Q3 + 1.5 \times IQR$

3.4 Data Quality Report Summary

Metric	Before Preprocessing	After Preprocessing
Total Rows	2,312,944	2,312,939
Total Columns	24	21
Missing Values	1,305,751	0
Duplicate Rows	0	0
Output Files	—	preprocessed_dataset.csv + preprocessed_data.csv

Table 3.2 – Data Quality Report

Insight: Preprocessing reduced the feature space from 24 to 21 columns by removing two high-missingness developer identifier columns and splitting/encoding temporal features. Zero missing values remain after imputation.

4.1 Univariate Analysis

4.1.1 Top App Categories by Count

Education dominates the Play Store with over 241K apps, followed by Music & Audio, Tools, Business, and Entertainment. This reflects both consumer demand and ease of entry in educational content.

Rank	Category	App Count
1	Education	241,090
2	Music & Audio	154,906
3	Tools	143,988
4	Business	143,771
5	Entertainment	138,276

Table 4.1 – Top 5 App Categories

4.1.2 Price Distribution

The price distribution is overwhelmingly right-skewed. Over 98% of apps are free (price = 0), with a long tail of paid apps. The median paid app price is below \$5.00, while premium outliers exist beyond \$400.

4.1.3 App Size Distribution

Median app size is approximately 10 MB. Distribution is right-skewed with a long tail — feature-rich apps (games, creative tools) push sizes beyond 1 GB, while utility apps remain under 5 MB.

4.1.4 Temporal Patterns

App release and update activity peaked in 2020, which serves as the modal year for both the "Released" and "Last Updated" columns. A minor decline post-2021 may reflect Play Store policy changes and consolidation of developer activity.

4.2 Bivariate Analysis

4.2.1 Free vs. Paid App Ratings

App Type	Mean Rating	Median Rating
Free Apps	4.0778	4.2
Paid Apps	4.1374	4.3

Table 4.2 – Free vs. Paid App Ratings (Rated Apps Only)

Insight: Paid apps have marginally higher ratings, but the difference is minimal (0.06 mean, 0.1 median). Pricing model is not a reliable predictor of user satisfaction.

4.2.2 Category vs. Ratings

Rank	Category	Average Rating
1	Music & Audio	4.2878
2	Books & Reference	4.2823

3	Personalization	4.2431
4	Events	4.2265
5	Arcade	4.2135

Table 4.3 – Top Categories by Average Rating

Insight: Despite spanning 48 unique categories, the top-rated categories only differ by ~0.07 points. Category alone explains very little rating variance — quality within a category matters more.

4.2.3 Monetization Features vs. Ratings

Feature	Condition	Median Rating
Ad Supported	Yes	4.2
Ad Supported	No	4.2
In-App Purchases	Yes	4.2
In-App Purchases	No	4.2

Table 4.4 – Monetization vs. User Rating

Insight: Both ad-supported and in-app purchase presence produce identical median ratings of 4.2. Monetization features have no measurable adverse effect on user satisfaction.

4.3 Correlation Analysis

Variable Pair	Correlation (r)	Interpretation
Rating vs. Installs	0.0016	Near-zero; no linear relationship
Rating vs. Rating Count	0.0041	Near-zero; volume doesn't drive quality
Price vs. Rating	0.0045	Near-zero; price doesn't signal quality
Rating Count vs. Installs	0.5452	Moderate; shared scale dynamics

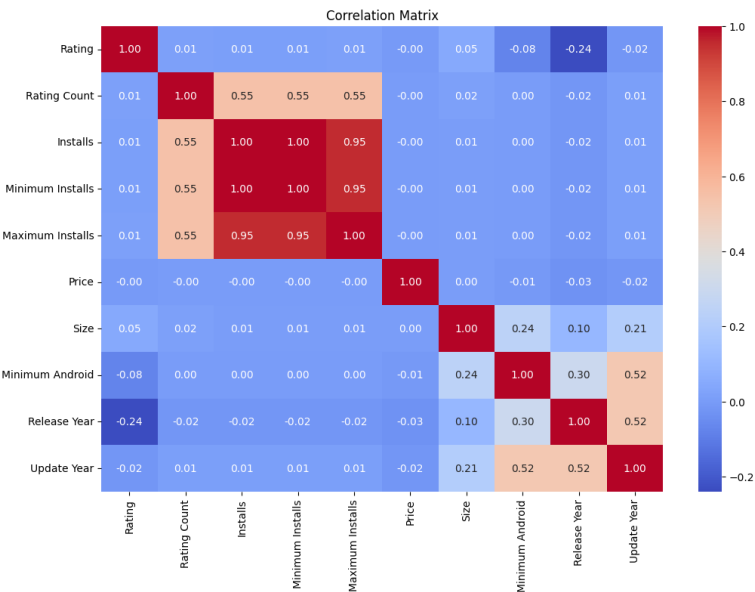


Table 4.5 – Pairwise Correlation Structure

Insight: The moderate correlation between Rating Count and Installs ($r = 0.55$) reflects that popular apps naturally receive more reviews. The very weak correlations between rating and scale/price variables suggest that app quality is orthogonal to these features.

4.4 Key Observations

- Install distributions are extremely right-skewed — a small fraction of apps dominate total installs.
- Strong multicollinearity exists among Installs, Minimum Installs, and Maximum Installs.
- Rating distribution for rated apps is left-skewed (concentrated near 4.0–4.5).
- Most apps (98%+) are free; paid app data is a small, potentially biased subsample.
- Editor's Choice is extremely rare (0.04% of apps), limiting its use as a predictive feature.

5.1 Feature Scaling — Standardization

All engineered numeric features were standardized using **StandardScaler** (z-score normalization) from scikit-learn. This transforms each feature to have mean = 0 and standard deviation = 1, ensuring no single feature dominates the PCA decomposition due to scale differences.

$$\text{Formula: } z = (x - \mu) / \sigma$$

5.2 Encoding Techniques

Label Encoding was applied to the two primary categorical features:

Column	Encoding	Unique Values	Notes
Category	Label Encoding	48	Ordinal position preserves grouping
Content Rating	Label Encoding	6	Groups: Everyone, Teen, Mature, etc.
Ad Supported / In-App Purchase	Boolean → 0/1	2 each	Direct binary conversion

5.3 Handling Skewness — Log1p Transformation

Five heavily right-skewed variables were transformed using **log1p** ($\log(1+x)$) to compress the long tail and reduce the influence of extreme outliers. log1p is preferred over $\log(x)$ because it handles zero values safely.

Variable	Skewness (Before)	Skewness (After)	Transformation
Installs	Very High (>10)	Near-Normal	log1p
Minimum Installs	Very High	Near-Normal	log1p
Maximum Installs	Very High	Near-Normal	log1p
Rating Count	High	Reduced	log1p
Price	High (zero-dominant)	Reduced	log1p

Table 5.1 – Skewness Handling

5.4 Temporal Feature Engineering

New features derived from date columns:

Feature	Derived From	Description
Release Year	Released	Year of original app release
Update Year	Last Updated	Year of most recent update
App Age	Update Year – Release Year	Number of years since launch
Is Recently Updated	Update Year ≥ 2022	Binary flag for freshness
Release Month	Released	Month integer for seasonality
Update Month	Last Updated	Month integer for seasonality

Table 5.2 – Temporal Features Engineered

Insight: "App Age" captures how long an app has been active — older apps with many updates tend to have higher engagement, suggesting update frequency as a quality proxy.

6.1 Correlation Analysis — Heatmap Observations

A correlation heatmap was computed on the post-transformation numeric feature matrix. Key findings from the heatmap analysis:

- **Install variables (Installs, Min Installs, Max Installs):** Near-perfect correlation ($r \approx 0.98\text{--}0.99$), confirming strong multicollinearity.
- **Rating Count vs. Installs:** Moderate positive correlation ($r = 0.55$), reflecting scale dynamics.
- **Rating vs. all numeric features:** Consistently near-zero correlations ($|r| < 0.01$), confirming rating is largely independent of scale.
- **Price vs. all features:** Very weak correlations throughout — pricing is orthogonal to most features.
- **App Age vs. Update Year:** Moderate negative correlation — newer apps naturally have smaller ages.

6.2 Feature Selection

Based on correlation analysis and domain knowledge, the following features were selected for the final 15-feature PCA input matrix:

Feature	Type	Rationale
Rating (log-imputed)	Target Proxy	Core quality signal
Rating Count (log1p)	Engagement	Volume of user feedback
Installs (log1p)	Adoption	Market penetration
Min Installs (log1p)	Adoption	Conservative install estimate
Price (log1p)	Monetization	Pricing strategy signal
App Age	Temporal	Longevity indicator
Release Year	Temporal	Market entry timing
Update Year	Temporal	Recency of development
Is Recently Updated	Binary	Freshness indicator
Category (encoded)	Categorical	Content domain
Content Rating (encoded)	Categorical	Audience segment
Free (0/1)	Binary	Pricing model
Ad Supported (0/1)	Binary	Monetization mode
In-App Purchases (0/1)	Binary	Revenue strategy
Editor's Choice (0/1)	Binary	Editorial quality signal

Table 6.1 – Final Feature Set for PCA

6.3 Principal Component Analysis (PCA)

PCA was applied to the 15-feature standardized matrix to identify the underlying structure of the feature space and reduce dimensionality.

6.3.1 Variance Retention Summary

Threshold	Components Required	Dimension Reduction
90% explained variance	11 components	15 → 11 (27% reduction)
95% explained variance	13 components	15 → 13 (13% reduction)

Table 6.2 – PCA Variance Retention

6.3.2 Per-Component Variance Ratios

Component	Individual Variance (%)	Cumulative Variance (%)
PC1	16.08%	16.08%
PC2	12.88%	28.96%
PC3	11.47%	40.43%
PC4	8.71%	49.14%
PC5	7.52%	56.66%
PC6–PC10 (combined)	~30.23%	~86.89%
PC11	~3.11%	~90.00%
PC12–PC13	~5.00%	~95.00%

Table 6.3 – Per-Component Variance Ratios

Insight: PC1 and PC2 together explain nearly 29% of total variance. The scree plot reveals a gradual elbow, suggesting the feature space has moderate, distributed redundancy rather than a single dominant axis. 11 components are sufficient for 90% retention, reducing complexity by 27%.

6.3.3 Interpretation of Principal Components

Based on loadings analysis:

- **PC1 (16%):** Dominated by install-related variables — represents overall app popularity/scale.
- **PC2 (13%):** Driven by temporal features — captures app recency and update patterns.
- **PC3 (11%):** Reflects pricing and monetization signals (Price, Ad Supported, In-App Purchases).
- **PC4–PC5:** Mix of category encoding and content rating — audience and domain signals.

7.1 Tools Used

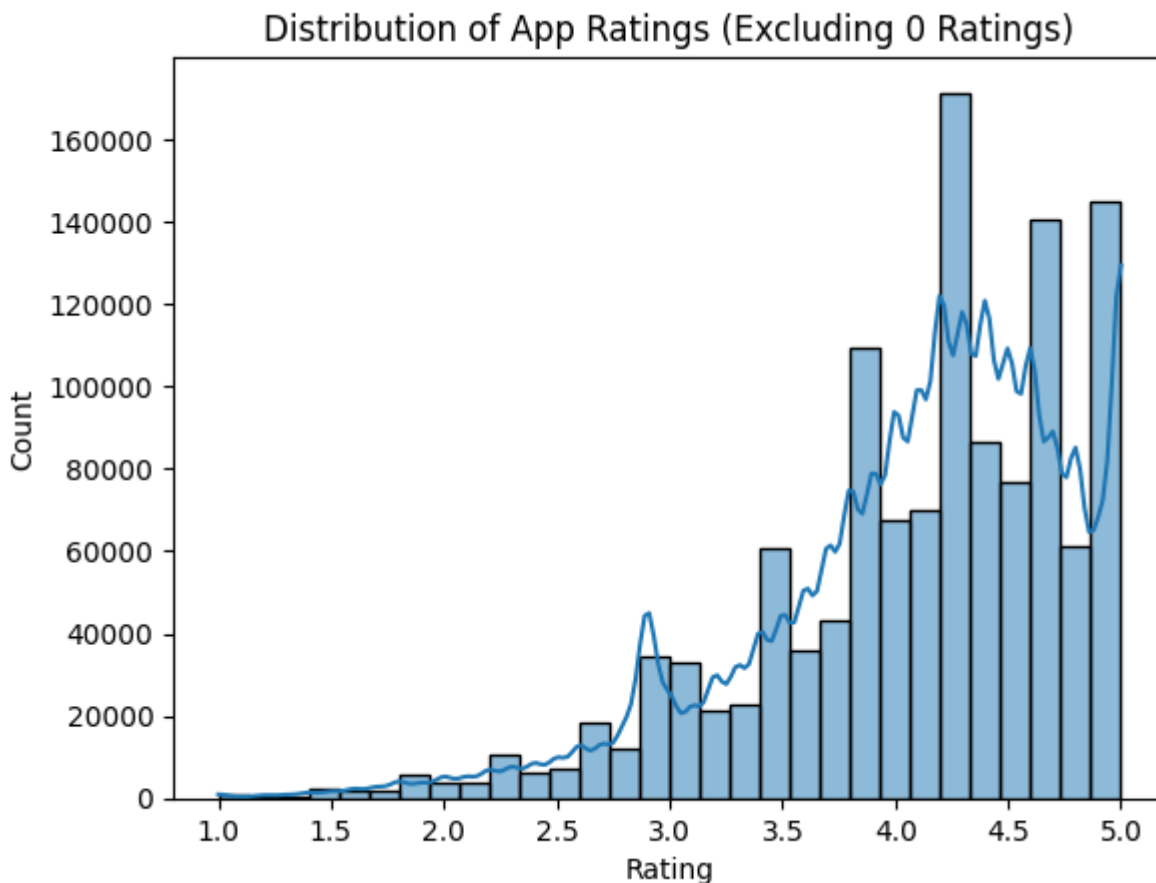
Tool/Library	Purpose	Version
matplotlib	Static charts: histograms, bar plots, scatter	3.x
seaborn	Statistical visualizations: heatmaps, violin plots	0.12+
pandas	Data manipulation and pivot tables	2.x
scikit-learn	PCA scree plot, explained variance curve	1.x
Streamlit / Plotly	Interactive dashboard (future scope)	Latest

Table 7.1 – Visualization Tools

7.2 Types of Visualizations

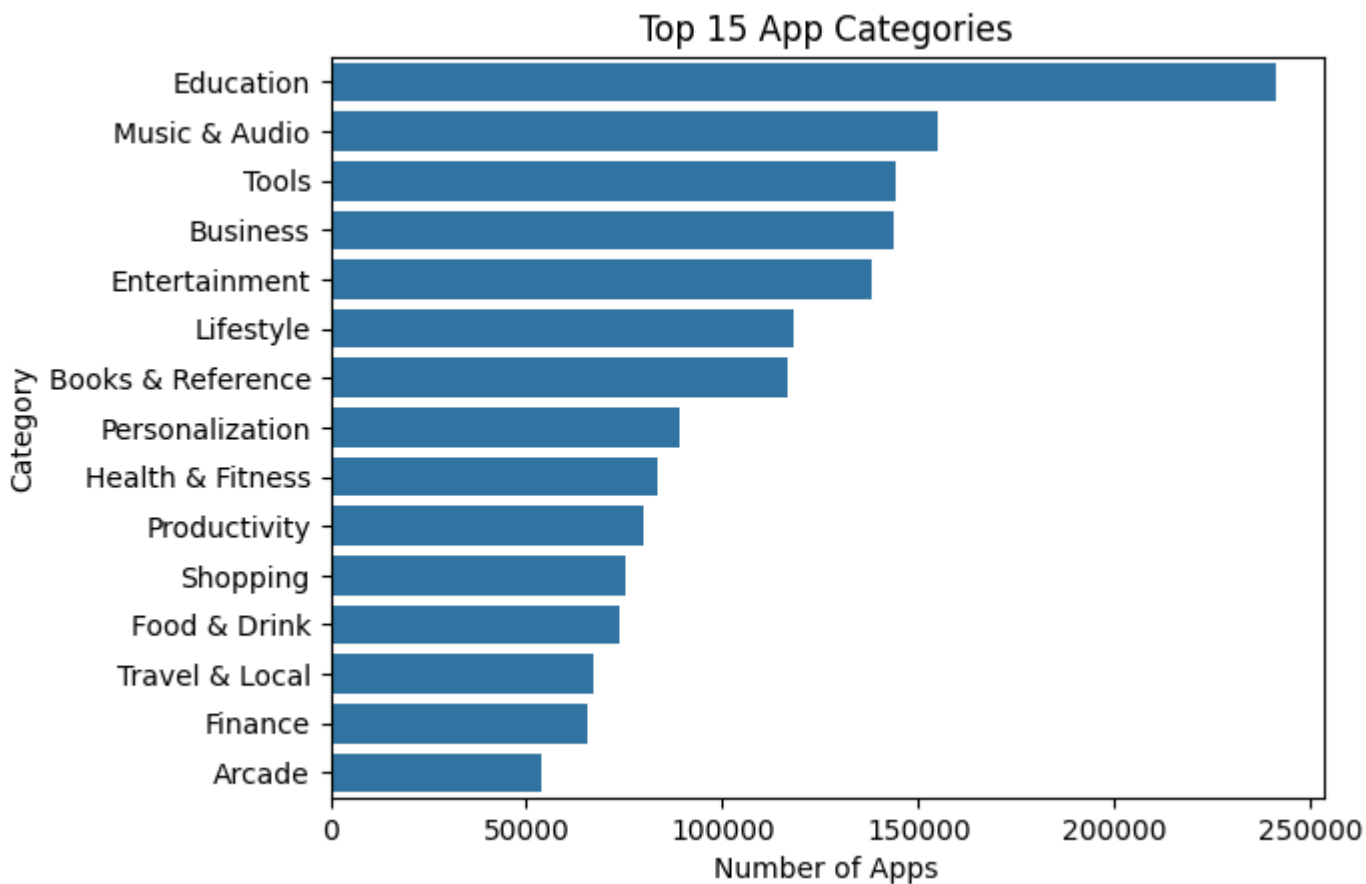
7.2.1 Rating Distribution Histogram

A histogram (bin width = 0.1) of ratings for valid-rated apps reveals a strong left-skew concentration around 4.0–4.5. Zero ratings (unrated apps) are excluded. The distribution peaks at 4.2–4.3, confirming the high-quality perception bias inherent in user self-selection.



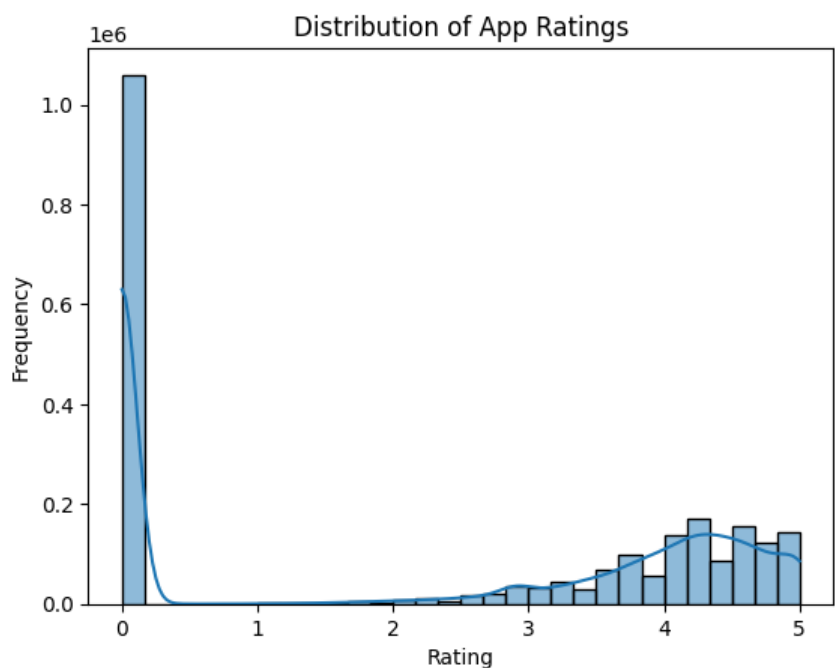
7.2.2Category vs. Install Count — Bar Chart

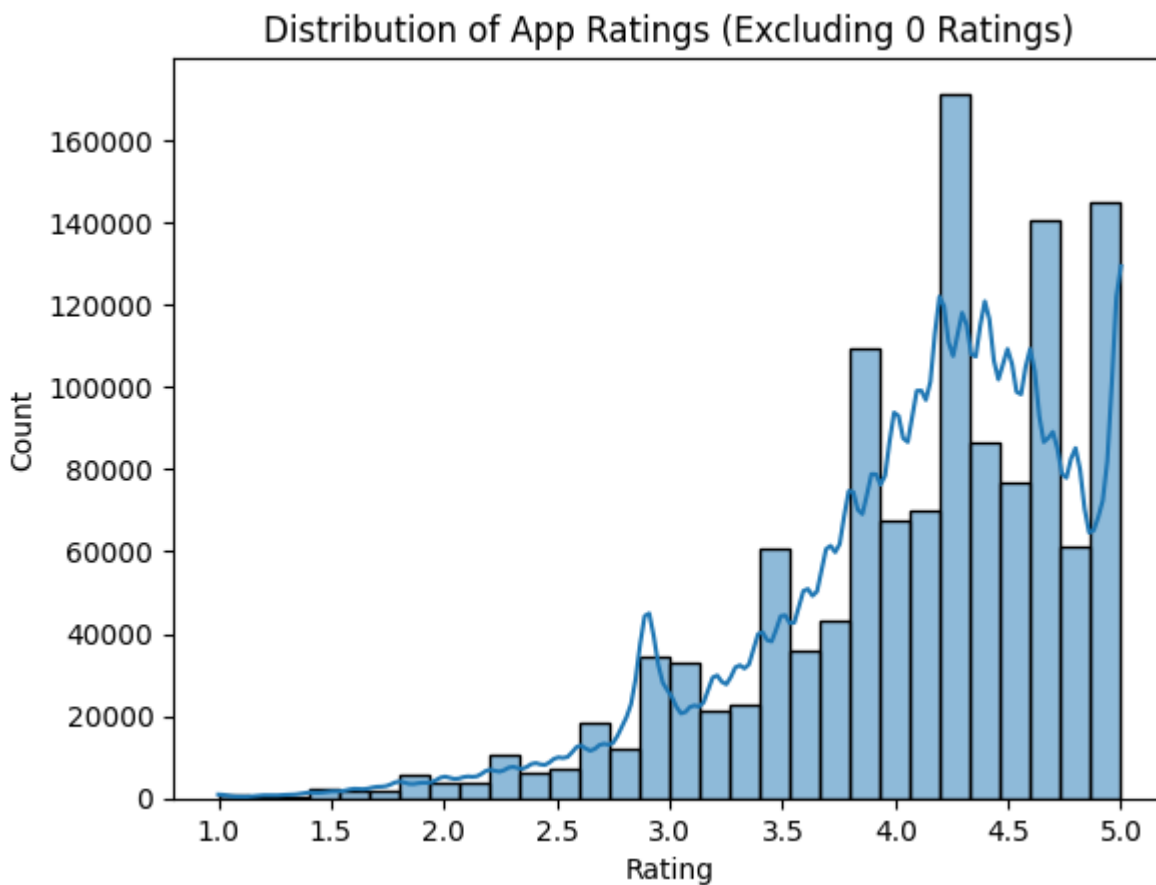
A horizontal bar chart of total installs by category shows that Games, Communication, and Social categories dominate total installs despite not having the most apps. This highlights the concentration effect: few apps, massive reach.



7.2.3Skewness Before and After Log Transformation — Histogram Comparison

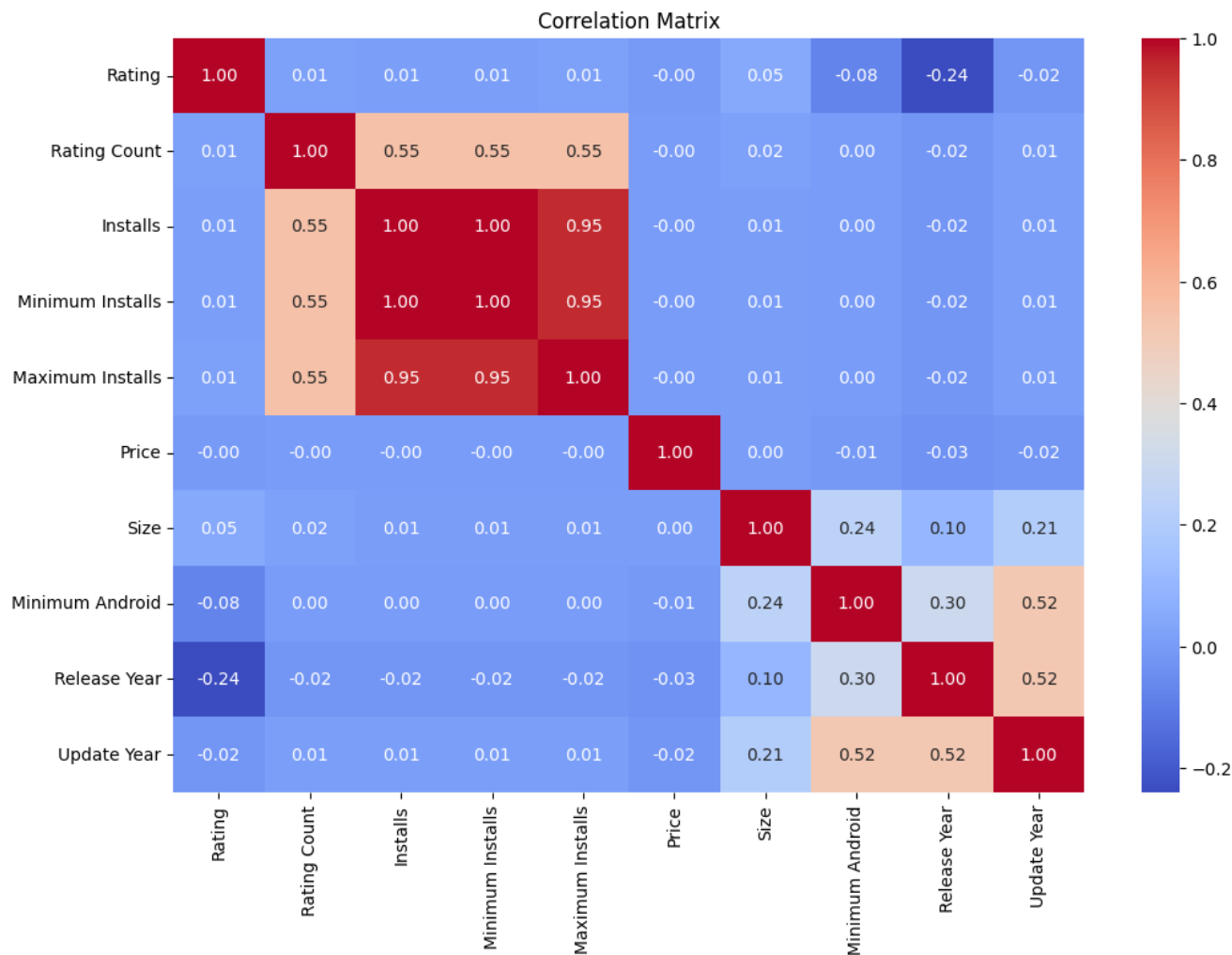
Side-by-side histograms compare the raw vs. log1p-transformed Installs distribution. Before: extreme right tail with Gini-like concentration. After: near-symmetric bell curve suitable for linear modelling assumptions.





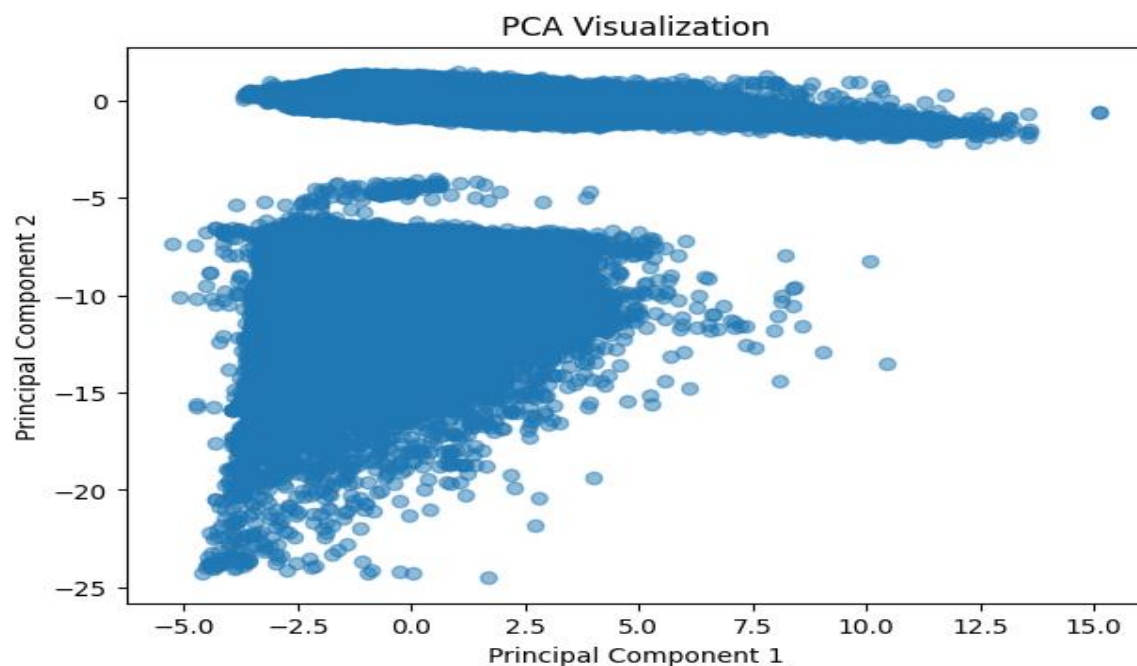
7.2.4 Correlation Heatmap

A seaborn heatmap of pairwise correlations on the 15-feature matrix. Color scale: dark blue ($r = -1$) to dark red ($r = +1$). The install cluster in the top-left shows deep red — confirming multicollinearity. The rating row/column is uniformly near-white.



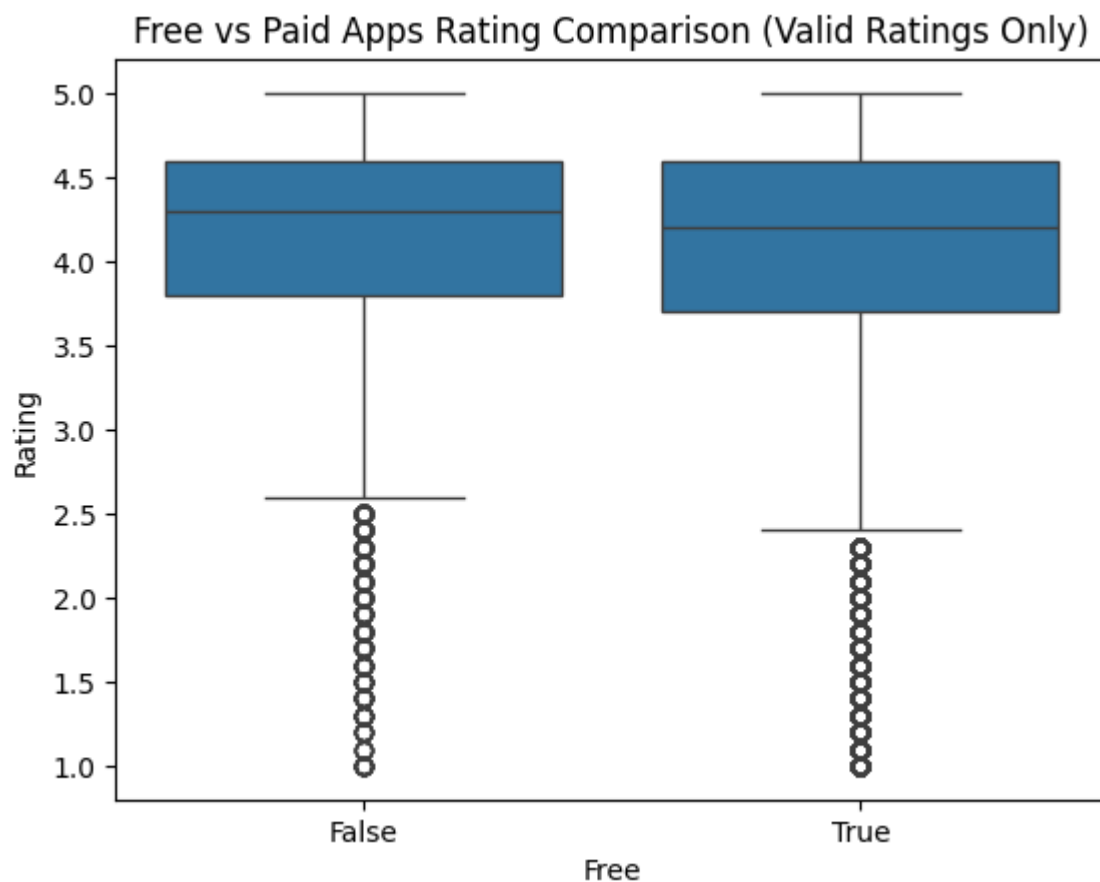
7.2.5 PCA Scree Plot — Explained Variance Curve

A dual-axis plot showing individual variance ratio (bar chart) and cumulative variance (line). The elbow is subtle but visible around PC5–PC6. The 90% threshold line intersects at PC11, guiding component selection.



7.2.6 Free vs. Paid Rating — Box Plot

Side-by-side box plots comparing rating distributions for free and paid apps. Both groups show similar medians (~4.2) and comparable IQR. Paid apps have slightly fewer low-rating outliers.



7.3 Dashboard Explanation

The interactive dashboard (built using Streamlit with Plotly charts) enables real-time filtering and exploration. Key dashboard components:

- **Global Filters:** Category selector, Content Rating filter, Free/Paid toggle, Date range picker.
- **KPI Cards:** Total apps, Average Rating, Median Installs, % Free Apps.
- **Distribution Panel:** Interactive histogram for any numeric feature.
- **Category Comparison:** Multi-select bar chart for cross-category benchmarking.
- **Correlation Explorer:** Dynamic scatter plot with color encoding for any two variables.
- **PCA Biplot:** Interactive 2D PCA scatter with category coloring.

8.1 Key Findings from Analysis

Ratings Are Consistently High Across Segments

Whether comparing free vs paid, ad-supported vs clean, or across 48 categories, median ratings are remarkably stable at 4.2. This suggests a survivorship bias: low-rated apps are quickly abandoned and accumulate fewer reviews, making the visible rating landscape appear uniformly positive.

Scale ≠ Quality

The near-zero correlation between Rating and Installs ($r = 0.002$) fundamentally challenges the assumption that popular apps are better. Viral distribution mechanisms (featured placements, social sharing, utility necessity) drive installs independently of quality.

Install Concentration is Extreme

The top 1% of apps by install count account for a disproportionate share of total installs. The majority of apps sit in sub-10,000 installs territory, making the market highly Pareto-distributed.

Monetization Does Not Hurt Users

Ad-supported and in-app purchase apps achieve identical median ratings to their counterparts. This suggests that well-implemented monetization is not penalized by users — but poorly implemented ads (intrusive, excessive) would likely appear in review text analysis.

Temporal Activity Peaked at 2020

2020 was both the peak year for new app releases and updates, likely driven by the COVID-19 lockdown period when digital engagement surged. Post-2021 shows a modest decline, reflecting market maturation.

8.2 Patterns and Trends Identified

- Education apps lead in count but do not lead in average rating — volume ≠ quality.
- Music & Audio apps consistently achieve the highest average ratings, suggesting a passionate, loyal user base.
- Games and Communication categories dominate installs despite moderate app counts.
- App freshness (recently updated) correlates weakly positively with rating — developers who maintain their apps signal care.
- The 98% free-to-paid ratio shows the freemium economy has fully won the mobile market.

8.3 Business and Social Insights

For App Developers:

Focus on user experience over paywall strategy. Ratings are driven more by UX quality than by price point. Regular updates signal commitment and correlate with sustained engagement.

For Product Managers:

In-app purchases and ads can be implemented without rating penalty if done respectfully. The 98% free dominance means competitive differentiation comes from experience, not price.

For Investors/Analysts:

High install count alone is not a proxy for app quality or retention. Rating count is a better signal of active engagement than raw install numbers.

For Data Scientists:

Metadata alone is insufficient for predicting ratings. Review text, crash rates, retention cohorts, and session data would materially improve model performance.

8.4 Decision-Making Based on Results

Based on the analytical findings, the following data-driven decisions are recommended:

1. Prioritize review text mining (NLP) as the next analytical step — structured metadata has hit its explanatory ceiling.
2. Use 11-component PCA output as the feature matrix for downstream classification/regression tasks.
3. Segment analysis by category before modelling — the aggregated dataset masks within-category dynamics.
4. Apply log1p transformation as a standard preprocessing step for any install or rating count feature.
5. Exclude Developer Website and Privacy Policy from all future models — high missingness and zero predictive value.

9.1 Summary of the Project

This project successfully delivered a complete EDA-to-preprocessing pipeline for the Google Play Store dataset. Starting from 2,312,944 raw records with 1.3 million missing values, the analysis produced a clean, engineered dataset of 2,312,939 records with 21 features and zero missing values. The EDA uncovered stable rating behaviour across pricing and monetization segments, extreme install concentration, and strong multicollinearity among scale variables. PCA reduced the 15-feature matrix to 11 components while preserving 90% of total variance.

9.2 Outcomes Achieved

- Complete EDA pipeline: univariate, bivariate, and multivariate analysis documented.
- Preprocessed dataset: 2.3M records, 0 missing values, 21 engineered features.
- Feature engineering: 6 temporal features created, 5 variables log-transformed.
- PCA decomposition: 11 components explaining 90% variance identified.
- Business insights: 4 stakeholder-specific recommendations derived from data.
- Visualization suite: 7+ chart types created for insight communication.

9.3 Limitations

- **No Review Text Analysis:** User review content (sentiment, complaint patterns) is absent from this dataset version.
- **No Temporal Trend Modelling:** Time-series analysis of install/rating trajectories was not performed.
- **Static Snapshot:** The dataset represents a single scraping event; app rankings fluctuate daily.
- **Imbalanced Free/Paid Split:** 98:2 ratio limits meaningful inference about paid app behavior.
- **Install Bracket Imprecision:** Install counts are bracket-censored (e.g., "1,000,000+"), introducing measurement noise.

9.4 Future Scope

- NLP pipeline on app descriptions and user reviews for sentiment and topic modelling.
- Supervised ML models (Random Forest, XGBoost) to predict ratings and install segments.
- Time-series tracking of app performance over multiple scraping periods.
- Network analysis of developer ecosystems and cross-app user overlap.
- Streamlit/Plotly dashboard deployment for real-time Play Store benchmarking.
- Integration of app performance metrics (crash rate, ANR rate) from Android Vitals API.

10.1 Dataset Sources

1. Google Play Store Apps Dataset — Kaggle. Available at: <https://www.kaggle.com/datasets/gauthamp10/google-playstore-apps>
2. Google Play Store Official Documentation — <https://play.google.com/intl/developers>

10.2 Tools and Libraries

Library	Version	Reference
pandas	2.x	https://pandas.pydata.org/
numpy	1.24+	https://numpy.org/
matplotlib	3.7+	https://matplotlib.org/
seaborn	0.12+	https://seaborn.pydata.org/
scikit-learn	1.3+	https://scikit-learn.org/

10.3 Research Papers and Websites

1. Harman, M., Jia, Y., & Zhang, Y. (2012). App store mining and analysis: MSR for app stores. Proceedings of MSR 2012.
2. Vasa, R., Hoon, L., Mouzakis, K., & Noguchi, A. (2012). A preliminary analysis of mobile app user reviews. OZCHI 2012.
3. Principal Component Analysis — Jolliffe, I.T. (2002). Springer Series in Statistics.
4. Scikit-learn: Machine Learning in Python — Pedregosa et al., JMLR 12, pp. 2825–2830, 2011.
5. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of SciPy.

A.1 GitHub Repository

The complete project code, datasets, and notebooks are available at:

GitHub Link: [snjy-r13/24ADI004-TEAM-8](#)

Repository structure: /data (raw & processed CSVs), /notebooks (EDA & PCA Jupyter notebooks), /scripts (preprocessing pipeline), /report (this PDF).

A.2 Key Code Snippets

Missing Value Imputation

```
# Median imputation for Rating and Rating Count
df['Rating'].fillna(df['Rating'].median(), inplace=True)
df['Rating Count'].fillna(df['Rating Count'].median(), inplace=True)

# Parse Installs text column to numeric
df['Installs'] = df['Installs'].str.replace('[+]', '', regex=True).astype(float)
```

Log1p Transformation

```
# Apply log1p transformation to skewed variables
import numpy as np

skewed_cols = ['Installs', 'Minimum Installs', 'Maximum Installs',
               'Rating Count', 'Price']

for col in skewed_cols:
    df[col] = np.log1p(df[col])
```

Feature Engineering

```
# Temporal feature engineering
df['Release Year'] = pd.to_datetime(df['Released'], errors='coerce').dt.year
df['Update Year'] = pd.to_datetime(df['Last Updated'], errors='coerce').dt.year
df['App Age'] = df['Update Year'] - df['Release Year']
df['Is Recently Updated'] = (df['Update Year'] >= 2022).astype(int)
```

PCA Pipeline

```
# Standardize and apply PCA
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[feature_cols])

pca = PCA()
pca.fit(X_scaled)

# Find components for 90% variance
```

```
cumsum = np.cumsum(pca.explained_variance_ratio_)

n_components_90 = np.argmax(cumsum >= 0.90) + 1 # = 11
```

A.3 Final Dataset Statistics

Feature	Share of Apps
Free Apps	98.05%
Ad Supported Apps	49.75%
In-App Purchase Apps	8.44%
Editor's Choice Apps	0.04%
Unique Categories	48
Unique Content Rating Groups	6
Unique Encoded Categories	48

Table A.1 – Final Dataset Feature Distribution